

TEST REPORT

Applicant/Manufacturer: Dragino Technology Co., Limited
Address: Room 202, BaoChengTai industrial park, No.8 CaiYun
LongCheng Street, LongGang District, Shenzhen China
Report Number: 2501X31094E-RFA1

Test Standard (s)

EN IEC 62311:2020; EN 50665:2017

Sample Description

Product Type: LoRaWAN Sensor Node
Model No.: DDS75-LB
Multiple Model(s) No.: please refer to Product Description for Equipment under
Test (EUT)
Trade Mark: DRAGINO
Date Received: 2025/09/26
Issue Date: 2026/01/04

Test Result:	Pass▲
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▲ In the configuration tested, the EUT complied with the standards above.

Prepared and Checked By:

Ekko Wu

Ekko Wu
RF Engineer

Approved By:

Nancy Wang

Nancy Wang
RF Supervisor



Note: The information marked # is provided by the applicant, the laboratory is not responsible for its authenticity and this information can affect the validity of the result in the test report. Customer model name, addresses, names, trademarks etc. are included.
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Bay Area Compliance Laboratories Corp. (Shenzhen)

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DOCUMENT REVISION HISTORY

Revision Number	Report Number	Description of Revision	Date of Revision
0	SZ1230802-44707E-RF	Original Report	2023-09-04
1	2501X31094E-RFA1	Amended Report	2026-01-04

Note:

This is an amended report based on the report SZ1230802-44707E-RF, the details as below:

(1) Adding the model number “LDS25-LB, LDS25-LS, LDS25-LB2, CS01-LB, CS01-LS, CS01-LB2, SE01-LS, SE01-LB2, DDS75-LS, DDS75-LB2, DDS45-LS, DDS45-LB2, DDS20-LS, DDS20-LB2, MDS120-LS, MDS120-LB2, SPH01-LS, SPH01-LB2, LMS01-LS, LMS01-LB2, WQS-LB, WQS-LS, WQS-LB2, SE0X-LB, SE0X-LS, SE0X-LB2, WSC2-LB, WSC2-LS, WSC2-LB2, MDS200-LS, MDS200-LB2, LDS40-LB, LDS40-LS, LDS40-LB2, LDS12-LS, LDS12-LB2, DDS04-LS, DDS04-LB2, RS485-LB2, RS485-LS, RS485W-LB, RS485W-LS, RS485W-LB2”, with the different power module or battery from the original model.

Based on above difference listed, the modifications will not impact any test data, we only updated the related photos in the report, all the test data and other photos please refer to the report SZ1230802-44707E-RF.

GENERAL INFORMATION

Product Description for Equipment under Test (EUT)

Product	LoRaWAN Sensor Node
Tested Model	DDS75-LB
Multiple Model(s)	LDS25-LB, LDS25-LS, LDS25-LB2, CS01-LB, CS01-LS, CS01-LB2, SE01-LB, SE01-LS, SE01-LB2, DDS75-LS, DDS75-LB2, DDS45-LB, DDS45-LS, DDS45-LB2, DDS20-LB, DDS20-LS, DDS20-LB2, MDS120-LB, MDS120-LS, MDS120-LB2, SPH01-LB, SPH01-LS, SPH01-LB2, LMS01-LB, LMS01-LS, LMS01-LB2, WQS-LB, WQS-LS, WQS-LB2, SE0X-LB, SE0X-LS, SE0X-LB2, WSC2-LB, WSC2-LS, WSC2-LB2, MDS200-LB, MDS200-LS, MDS200-LB2, LDS40-LB, LDS40-LS, LDS40-LB2, LDS12-LB, LDS12-LS, LDS12-LB2, DDS04-LB, DDS04-LS, DDS04-LB2, RS485-LB, RS485-LB2, RS485-LS, RS485W-LB, RS485W-LS, RS485W-LB2
Frequency Range	BLE: 2402-2480MHz SRD: 865-868MHz & 868-868.6MHz
Maximum Transmit Power (EIRP)	BLE: -0.06dBm SRD: 13.17dBm
Antenna Specification [#]	2dBi(It is provided by the manufacturer)
Voltage Range	LB Series: DC 3.6V from battery 1/4 LB2 Series: DC 3.6V from battery 3 LS Series: DC 3.7V from battery 2
Sample/EUT Status	Good condition
Adapter Information	N/A
Battery Information	Original: Battery 1(EVE ENERGY): DC 3.6V, 8500mAh New: Battery 2(YO): DC 3.7V, 3000mAh 11.1Wh Battery 3(RAMWAY): DC 3.6V, 19000mAh Battery 4(RAMWAY): DC 3.6V, 8500mAh
Note: The Multiple models are electrically identical with the test model except for model no., probes, PCB and battery. Please refer to the declaration letter [#] for more detail, which was provided by manufacturer.	

EXHIBIT A - EUT PHOTOGRAPHS

Please refer to the report number is 2501X31094E-EUT.

EXHIBIT B - PRODUCT SIMILARITY DECLARATION OF LETTER

Dragino Technology Co., Limited.
Address: Room 202,BaoChengTai industrial park,No.8 CaiYun LongCheng Street,LongGang District,
Shenzhen China
Tel: 86-0755-86610821; Fax: 86-0755-61673660
Email: edwin@dragino.com

Product Similarity Declaration

Date: 2025-11-09

Dear Sir/Madam,

We, **Dragino Technology Co., Limited.** hereby declare that we have a product named as LoRaWAN Sensor Node (model: DDS75-LB),were tested by Bay Area Compliance Laboratories Corp. (Shenzhen), Meanwhile, For our marketing purpose, we would like to list a series model(LDS25-LB, LDS25-LS, LDS25-LB2, CS01-LB, CS01-LS, CS01-LB2, SE01-LB, SE01-LS, SE01-LB2, DDS75-LB, DDS75-LS, DDS75-LB2, DDS45-LB, DDS45-LS, DDS45-LB2, DDS20-LB, DDS20-LS, DDS20-LB2, MDS120-LB, MDS120-LS, MDS120-LB2, SPH01-LB, SPH01-LS, SPH01-LB2, LMS01-LB, LMS01-LS, LMS01-LB2, WQS-LB, WQS-LS, WQS-LB2, SE0X-LB, SE0X-LS, SE0X-LB2, WSC2-LB, WSC2-LS, WSC2-LB2, MDS200-LB, MDS200-LS, MDS200-LB2, LDS40-LB, LDS40-LS, LDS40-LB2, LDS12-LB, LDS12-LS, LDS12-LB2, DDS04-LB, DDS04-LS,DDS04-LB2, RS485-LB, RS485-LB2, RS485-LS, RS485W-LB, RS485W-LS, RS485W-LB2) on reports and certificate. All the models are identical in electrically, the differences are shown in the table below. No other changes were made to them.

Model	Probes	PCB	Battery
DDS75-LB	Ultrasonic Probe	without charging module	8500mAh battery
RS485-LB	N/A		
SE01-LB	Soil Moisture Probe		
DDS45-LB	Ultrasonic Probe		
DDS20-LB	Ultrasonic Probe		
MDS120-LB	Microwave Probe		
MDS200-LB	Microwave Probe		
LDS12-LB	LiDar Probe		
SPH01-LB	Soil PH probe		
LMS01-LB	Leaf Moisture Probe		
DDS04-LB	Ultrasonic Probe		
LDS25-LB	LiDAR Distance Probe		
CS01-LB	Current Probe		
WQS-LB	Smoke Probe		
SE0X-LB	DR-SE-6P Soil Moisture & EC. Probe		
WSC2-LB	Alarm		
LDS40-LB	LiDAR Distance Probe		

DDS75-LS	Soil Moisture Probe	Has a charging module	3000mAh rechargeable battery
RS485-LS	without Probe		
SE01-LS	Ultrasonic Probe		
DDS45-LS	Microwave Probe		
DDS20-LS	Microwave Probe		
MDS120-LS	LiDar Probe		
MDS200-LS	Soil PH probe		
LDS12-LS	Leaf Moisture Probe		
SPH01-LS	Ultrasonic Probe		
LMS01-LS	Soil Moisture Probe		
DDS04-LS	Ultrasonic Probe		
LDS25-LS	LiDAR Distance Probe		
CS01-LS	Current Probe		
WQS-LS	Smoke Probe		
SE0X-LS	DR-SE-6P Soil Moisture & EC. Probe		
WSC2-LS	Alarm		
LDS40-LS	LiDAR Distance Probe		
DDS75-LB2	Soil Moisture Probe	Without charging module	19000mAh battery
RS485-LB2	without Probe		
SE01-LB2	Ultrasonic Probe		
DDS45-LB2	Microwave Probe		
DDS20-LB2	Microwave Probe		
MDS120-LB2	LiDar Probe		
MDS200-LB2	Soil PH probe		
LDS12-LB2	Leaf Moisture Probe		
SPH01-LB2	Ultrasonic Probe		
LMS01-LB2	Soil Moisture Probe		
DDS04-LB2	Ultrasonic Probe		
LDS25-LB2	LiDAR Distance Probe		
CS01-LB2	Current Probe		
WQS-LB2	Smoke Probe		
SE0X-LB2	DR-SE-6P Soil Moisture & EC. Probe		
WSC2-LB2	Alarm		
LDS40-LB2	LiDAR Distance Probe		
RS485W-LS	Without probe	1. Has a charging module 2. Support	3000mAh rechargeable battery

		12Vdc output	
RS485W-LB	Without probe	3. Has a charging module 4. Support 12Vdc output	8500mAh battery
RS485W-LB2	Without probe	1. Has a charging module 2. Support 12Vdc output	19000mAh battery

We confirm that all information above is true, and we'll be responsible for all the consequences. Please contact me if you have any question.

Sincerely Yours,

Signature

edwin chen

Manager



BELOW IS THE REFERENCED REPORT

TEST REPORT

Applicant/Manufacturer: Dragino Technology Co., Limited
Address : Room 202,BaoChengTai industrial park,No.8 CaiYun
LongCheng Street,LongGang District,Shenzhen China
Report Number : SZ1230802-44707E-RF

Test Standard (s)

EN IEC 62311:2020; EN 50665:2017

Sample Description

Product Type: LoRaWAN Sensor Node
Model No.: DDS75-LB
Multiple Model(s) No.: RS485-LB, SE01-LB, DDS45-LB, DDS20-LB, MDS120-LB,
MDS200-LB, LDS12-LB, SPH01-LB, LMS01-LB, DDS04-LB
Trade Mark: DRAGINO
Date Received: 2023/08/02
Report Date: 2023/09/04

Test Result:	Pass*
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* In the configuration tested, the EUT complied with the standards above.

Prepared and Checked By:

April Zhang

April Zhang
RF Engineer

Approved By:

Nancy Wang

Nancy Wang
RF Supervisor

Note: The information marked * is provided by the applicant, the laboratory is not responsible for its authenticity and this information can affect the validity of the result in the test report. Customer model name, addresses, names, trademarks etc. are included.

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Bay Area Compliance Laboratories Corp. (Shenzhen)

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DOCUMENT REVISION HISTORY

Revision Number	Report Number	Description of Revision	Date of Revision
0	SZ1230802-44707E-RF	Original Report	2023-09-04

GENERAL INFORMATION

Product Description for Equipment under Test (EUT)

Product	LoRaWAN Sensor Node
Test Model	DDS75-LB
Multiple Model(s)	RS485-LB, SE01-LB, DDS45-LB, DDS20-LB, MDS120-LB, MDS200-LB, LDS12-LB, SPH01-LB, LMS01-LB, DDS04-LB (Please refer to the DoS* provided by the applicant)
Frequency Range	BLE: 2402-2480MHz SRD: 865-868MHz & 868-868.6MHz
Maximum ERP/EIRP	BLE: -0.06dBm SRD: 13.17dBm
Antenna Specification*	2dBi(It is provided by the manufacturer)
Voltage Range	DC 3.6V from battery
Sample serial number	299I-1 (Assigned by BACL, Shenzhen)
Sample/EUT Status	Good condition

Objective

This test report is in accordance with EN IEC 62311:2020, Generic standard to demonstrate the compliance of electronic and electrical apparatus with the basic restrictions related to human exposure to electromagnetic fields (0 Hz–300 GHz) is to demonstrate the compliance of apparatus with the basic restrictions or reference levels on exposure of the general public related to electric, magnetic, electromagnetic fields as well as induced and contact current. It also in accordance with EN 50665:2017, Generic standard for assessment of electronic and electrical equipment related to human exposure restrictions for electromagnetic fields (0 Hz - 300 GHz).

The objective is to determine the compliance of EUT with EN IEC 62311:2020 & EN 50665:2017.

EUT Exercise Software

No exercise software was used.

Test Facility

The Test site used by Bay Area Compliance Laboratories Corp. (Shenzhen) to collect test data is located on the 5F(B-West) , 6F, 7F, the 3rd Phase of Wan Li Industrial Building D, Shihua Rd, FuTian Free Trade Zone, Shenzhen, China.

Each test item follows test standards and with no deviation.

Technical Requirements Specification in EN IEC 62311:2020 & EN 50665:2017

Generic standard to demonstrate the compliance of electronic and electrical apparatus with the basic restrictions related to human exposure to electromagnetic fields (0 Hz–300 GHz) is to demonstrate the compliance of apparatus with the basic restrictions or reference levels on exposure of the general public related to electric, magnetic, electromagnetic fields as well as induced and contact current.

RF Exposure Evaluation

Limit:

According to EN IEC 62311:2020, the criteria listed in the below table shall be used to evaluate the environmental impact of human exposure to radio-frequency (RF) radiation as specified table 2 of Council Recommendation 1999/519/EC.

Reference levels for electric, magnetic and electromagnetic fields
(0 Hz to 300 GHz, unperturbed rms values)

Frequency range	E-field strength (V/m)	H-field strength (A/m)	B-field (μT)	Equivalent plane wave power density S_{eq} (W/m ²)
0-1 Hz	—	$3,2 \times 10^4$	4×10^4	—
1-8 Hz	10 000	$3,2 \times 10^4/f^2$	$4 \times 10^4/f^2$	—
8-25 Hz	10 000	$4\,000/f$	$5\,000/f$	—
0,025-0,8 kHz	$250/f$	$4/f$	$5/f$	—
0,8-3 kHz	$250/f$	5	6,25	—
3-150 kHz	87	5	6,25	—
0,15-1 MHz	87	$0,73/f$	$0,92/f$	—
1-10 MHz	$87/f^{1/2}$	$0,73/f$	$0,92/f$	—
10-400 MHz	28	$0,073$	$0,092$	2
400-2 000 MHz	$1,375\, f^{1/2}$	$0,0037\, f^{1/2}$	$0,0046\, f^{1/2}$	$f/200$
2-300 GHz	61	0,16	0,20	10

Notes:

1. f as indicated in the frequency range column.

Test method**Far Field Calculation Formula**

$$E = \frac{\sqrt{30PG(\theta, \phi)}}{r}$$

G = antenna gain relative to an isotropic antenna
 θ, ϕ = elevation and azimuth angles to point of investigation
 r = distance from observation point to the antenna

Evaluated Data

For Worst case:

Mode	Frequency Range (MHz)	Antenna Gain [*] (note 2)		Maximum Conducted Power (note 1) (dBm)	Target Power [*] (note 3)		Prediction Distance (m)	E-field strength (V/m)	Limit (V/m)
		Typical (dBi)	numeric		dBm	Watts			
BLE	2402-2480	2.0	1.58	-2.06	-2.0	0.0006	0.2	0.866	61.00
SRD	865-868.6	2.0	1.58	13.32	13.5	0.0224	0.2	5.159	40.44

Note 1: The detail test result and the test date please refer to RF test reports: SZ1230802-44707E-RF-22A/-22B.

Note 2: The antenna gain provided by the applicant.

Note 3: Data provided by the applicant, which not higher than test data 2dB.

Conclusion:

The RF Exposure is compliance.

EXHIBIT A - EUT PHOTOGRAPHS

Please refer to the report number is SZ1230802-44707E-EUT.

EXHIBIT B – PRODUCT SIMILARITY DECLARATION OF LETTER

Dragino Technology Co., Limited
Address: Room 202, BaoChengTai industrial park, No.8 CaiYun LongCheng Street,
LongGang District, Shenzhen China
Tel: 0755-86610821
Email: edwin@dragino.com

Product Similarity Declaration

Date: 2023-08-23

Dear Sir/Madam,

We, Dragino Technology Co., Limited hereby declare that we have a product named as LoRaWAN Sensor Node (model: DDS75-LB), were tested by BACL (Shenzhen). Meanwhile, For our marketing purpose, we would like to list a series model (RS485-LB, SE01-LB, DDS45-LB, DDS20-LB, MDS120-LB, MDS200-LB, LDS12-LB, SPH01-LB, LMS01-LB, DDS04-LB) on reports and certificate. All the models are identical in electrically, the differences are shown in the table below. No other changes were made to them.

Model	Probes
DDS75-LB	Ultrasonic Probe
RS485-LB	N/A
SE01-LB	Soil Moisture Probe
DDS45-LB	Ultrasonic Probe
DDS20-LB	Ultrasonic Probe
MDS120-LB	Microwave Probe
MDS200-LB	Microwave Probe
LDS12-LB	LiDar Probe
SPH01-LB	Soil PH probe
LMS01-LB	Leaf Moisture Probe
DDS04-LB	Ultrasonic Probe

We confirm that all information above is true, and we'll be responsible for all the consequences.
Please contact me if you have any question.

Sincerely Yours,

Signature:

Edwin
Chen
数字签名
者: Edwin Chen
日期: 2023.09.01
15:19:18 +08'00'

Edwin chen
CEO

***** END OF REPORT *****